

Rohit Sudhakar

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Education

University of Wisconsin – Madison Expected May 2026
Bachelor of Science in Computer Engineering: Machine Learning and Data Science track
GPA: 3.7/4.00 Dean's Honor List
Relevant Coursework: Digital System Fundamentals; Signals, Information, and Computation; Discrete Mathematics; Object-Oriented Programming; Python for Data Science and Machine Learning; Electronic Circuits; Machine Organization and Programming.

Work Experience

Ardent Capital Private Limited — AI Intern May 2025 – Present

- Developed an AI-based Banking Transaction Categorization Model processing 1.4M+ transactions; integrated fuzzy matching for rare/unseen categories.
- Built a React front-end that allows internal users to upload bank statements, view categorized transactions, analyze summaries, and trigger re-training of the AI model from the website.
- Automated internal workflows using Python, enhancing efficiency and accuracy across multiple tasks.
- Collaborated with cross-functional teams to integrate AI outputs into dashboards and reporting tools.

Jubilant Capital — Software Engineering Intern May 2024 – Aug 2024

- Led full-stack development of a client management system (React/Tailwind front-end, Flask/SQL backend), improving efficiency by 20%.
- Strengthened data security via role-based access control and encryption protocols.

- Optimized backend APIs and queries for 1,000+ client records, improving performance and reliability.

Connected & Autonomous Transportation Systems Lab Mar 2024
– May 2024
Undergraduate Researcher — Madison, WI

- Conducted research on connected and autonomous vehicles; analyzed real-time sensor data using C and MATLAB.
- Contributed to improvements in vehicle-to-vehicle communication and safety, achieving a 15% efficiency gain.

Infinity Credit Solutions — Software Engineering Intern May 2023
– Sept 2023

- Optimized internal workflows and automated processes using Python and API integrations, increasing productivity by 25%.
- Enhanced Zoho CRM features to meet company needs, improving efficiency and reporting.

Projects

Blackjack Game with FreeRTOS FSM (C — FreeRTOS — Embedded Systems — LCD — EEPROM — TRNG) Mar 2025 – May 2025

- Built a state-driven Blackjack game on a microcontroller using FreeRTOS with tasks for each game phase; implemented LCD UI, joystick/button input, EEPROM high-score storage, and TRNG-based shuffling.
- Managed inter-task communication with semaphores, queues, and event groups for real-time gameplay synchronization.
- Enabled persistent scoring and reliable game state management for repeated plays.

Client Database Management System (React — Tailwind — Flask — SQL — PyMongo) Aug 2024

- Developed a scalable client database system for 500+ clients with secure backend APIs and optimized SQL queries for 1,000+ records.
- Implemented role-based access control and bulk upload, reducing manual data entry by 60%.

- Designed an intuitive React/Tailwind front-end for internal teams, improving usability and navigation efficiency.

Document Automation Tool (Python — Tkinter — Docx) Jul 2023

- Automated generation of over 20 daily documents for banks/NBFCs, cutting processing time by 50%.
- Designed dynamic input templates and automated placeholder replacement in Word documents, reducing errors.

Graphical Earthquake Monitor and Pathfinder (Java — JavaFX — Dijkstra's Algorithm) Apr 2024

- Developed a real-time map application using JavaFX; empowered by Dijkstra's algorithm for efficient pathfinding, reducing route calculation time by 50%.
- Built an interactive interface that allows users to explore multiple earthquake scenarios and optimized routes for relief planning.

Technical Skills

Programming: Java, Python, Arduino-based C++, MATLAB, JavaScript, Assembly
 Frameworks/Libraries: JavaFX, Flask, React, Tailwind CSS, TensorFlow, Keras
 Databases: MySQL, SQL, MongoDB, Firebase
 Tools: Git, Docker, Kubernetes, Jupyter, VS Code, PyCharm
 Dev/Scripting: Bash/Command Line, Object-Oriented Programming, API Development